MIL-T-45317E
21 February 1973
SUPERSEDING
MIL-C-45317D
3 March 1967

MILITARY SPECIFICATION

TRUCK, PLATFORM UTILITY: 1/2 TON, 4x4 M274 SERIES

This specification is approved for use by the Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers lightweight utility vehicles for use by airborne and infantry units in combat areas.

1.2 Classification. Vehicles shall be of the following types as specified by the procuring activity (see 6.2):

Type I - Truck, Platform Utility: 1/2 Ton, 4X4, M274A2 (Steering control of the 2 front wheels or all 4 wheels).
Type II - Truck, Platform Utility: 1/2 Ton, 4X4, M274A5 (Steering control of the 2 front wheels only).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bid or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS
Military

MIL-L-2104	-	Lubricating Oil, Internal-Combustion Engine, Heavy-Duty.
MIL-L-2105	-	Lubricant Oil, Gear, Multipurpose.
MIL-G-3056	-	Gasoline, Automotive, Combat.
MIL-R-3065	-	Rubber, Fabricated Products.
MIL-A-8421	-	Air Transportability Requirements, General Specification For.
MIL-L-10324	-	Lubricant, Gear, Universal, Subzero.
MIL-G-10924	-	Grease, Automotive and Artillery.
MIL-L-21260	-	Lubricating Oil, Internal-Combustion Engine, Preservative.

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PURCHASE DESCRIPTION

DAPD-298 - Engine, Gasoline: 2 Cylinder, Horizontal Opposed Air Cooled, 14 HP, A042.

STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-130 - Identification, Marking of U. S. Military Property.

MIL-STD-193 - Painting Procedures Tactical Vehicles (Tracked and Wheeled).

MIL-STD-209 - Slinging Eyes and Attachments For Lifting and Tying Down Heavy Military Equipment.

MIL-STD-281 - Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.

MIL-STD-417 - Rubber Compositions, Vulcanized General Purpose, Solid (Symbols and Tests).

MIL-STD-461 - Electromagnetic Interference Characteristics, Requirements For Equipment.

MIL-STD-642 - Identification Marking of Administrative, Combat, and Tactical Transport Vehicles.

MIL-STD-882 - System Safety Program For Systems And Associated Subsystems And Equipment: Requirement For.

DRAWINGS

C10957514 - Vehicle Run-in Test Stand Performance Test Equipment Specifications.

C10944366 - Chart, Production Lubrication

PUBLICATIONS

Engineering Parts Lists

8736605 - Truck, Platform Utility: 1/2 Ton, 4 x 4, M274A2.

8736756 - Truck, Platform Utility: 1/2 Ton, 4 x 4, M274A5.

Master Lists

8736605 - Master List of Supplementary Quality Assurance Provisions (MSQAP), For Truck, Platform Utility: 1/2 Ton, 4 x 4 M274A2.

8736756 - Master List of Supplementary Quality Assurance Provisions (MSQAP), For Truck, Platform Utility: 1/2 Ton, 4 x 4, M274A5.

8736605 - Master List of Inspection Equipment Drawings for Truck, Platform Utility: 1/2 Ton, 4 x 4, M274A2.

8736756 - Master List of Inspection Equipment Drawings for Truck, Platform Utility: 1/2 Ton, 4 x 4, M274A5.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

8-6-66 - Human Engineering Laboratories (HEL)

(Application for copies should be addressed to U. S. Army Engineering Laboratories, Aberdeen Proving Ground, Maryland).

Federal Register, Volume 35 Number 21

(Application for copies should be addressed to Environmental Protection Agency, Office of the Administrator, Motor Vehicles, Washington, D. C.).

3. REQUIREMENTS

3.1 Preproduction model. When specified (see 6.2), a preproduction model(s) of the complete vehicle to be furnished under this specification shall be produced prior to the manufacture or fabrication of vehicles in quantity. The vehicle(s) when completed shall be submitted to the Government for examination and tests to determine conformance to the requirements of this specification. Vehicle(s) submitted by the contractor shall be fully representative of vehicles proposed to be supplied by the vehicle manufacturer from production facilities and tooling under the supply contract.

3.2 First production vehicle(s). The first production vehicle(s), shall be fully representative of vehicles proposed to be furnished under the contract, and shall be submitted to the Government for inspection to determine conformance to the requirements of this specification.

3.2.1 Reliability. The vehicle, when operated over terrain as described herein, shall possess a mean miles between failures (MMBF) of not less than 1500 miles. This value is based on a test of not less than 4000 miles.

(a)	Level cross country	30 percent
(b)	Hilly cross country	30 percent
(c)	Gravel and dirt roads	30 percent
(d)	Belgian block	2-1/2 percent
(e)	Paved roads	7-1/2 percent

A-failure for calculation of mean miles between failure (MMBF) shall be as defined by the joint CDC/AMC wheel and track vehicle mission failures definition, dated 27 March 1972.

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a. A "failure" is defined as any malfunction of the end item that requires corrective action which cannot be deferred:

(1) Until the next scheduled maintenance (exclusive of lubrication services) if organizational maintenance is prescribed for correction, or--

(2) For the remainder of its specified life before overhaul, replacement, rebuild or salvage (as applicable) if direct or general support maintenance is prescribed for correction.

b. Corrective action is not deferrable if the malfunction causes (or would cause if not corrected):

(1) Inability to commence operation, cessation of operation or degradation of performance capability of the end item below that required for the accomplishment of its prescribed mission function, or--

(2) A critical or catastrophic hazard to personnel or equipment as defined by MIL-STD-882.

c. Any malfunction which the operator or crew (user) can remedy will not be considered a failure provided that the repair is authorized or prescribed as an operator function and can be accomplished within 30 minutes using only controls, tools or spare parts incorporated in or carried with the end item.

3.2.2 Durability. The M274, with rated payload, shall have a .60 probability of completing a minimum of 4000 miles without replacement, rebuild, or overhaul of any of the following major components, i.e., engine, transmission, transfer or differential, and shall have a .90 probability of completing the first 4000 miles without repair or rebuild of body or frame.

3.2.3 Maintainability.

1. The total scheduled maintenance excluding driver/crew checks and services shall be between 8 and 11 manhours, and the total unscheduled maintenance shall be between 0 and 15 manhours during the first 4000 miles of operation. This equates to a Maintenance Ratio (MR) of .020 to .065 at an average speed of 10 mph and a (MR) of .04 to .13 at an average speed of 20 mph.

2. The schedules maintenance intervals shall be 6 months or 3000 miles of operation, whichever comes first.

3. Maximum time to repair (to include diagnosis, repair and verification) using personnel normally employed at:

a. Organizational maintenance shall not exceed 4 manhours 95% of the time.

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b. Direct support maintenance shall not exceed 8 manhours, 95% of the time.

c. General support maintenance, in a backup Direct Support role, shall not exceed 40 manhours 95% of the time.

3.2.4 Transportability. The vehicle shall be transportable by ship, landing vessels and railways without major disassembly. The vehicle shall be transportable in Phase III Air Borne operation and shall meet the requirements of MIL-A-8421.

3.3 Materials. Materials as specified herein and in the referenced drawings, specifications, and standards shall be free from all defects and imperfections that might affect the serviceability and function of the finished product.

3.3.1 Qualified products. With respect to all assemblies and component parts requiring product qualification in accordance with the pertinent specifications or drawings listed on the EPL cited in 3.4, and necessary in the construction of this vehicle, the contractor shall be responsible for using only those products which are listed (by part of drawing number as QPL items) on the QPL in the referenced specification or which have been approved for inclusion on such lists. Manufacturer's inspection records shall specifically list such component(s), name of supplier(s) (as listed on the QPL or in the approval letter), and number and date of the QPL from which the selection was made or date of approval letter when products are approved but not as yet listed on the QPL.

3.3.2 Ozone resistant. Rubber products procured under requirements of specification MIL-R-3065 are to be ozone resistant to the degree specified by the C_1 requirements of Standard MIL-STD-417. Rubber components controlled by the specification referenced on the drawing.

3.4 Construction. Vehicle, components, subassemblies, and assemblies shall be fabricated and assembled into a complete vehicle in accordance with drawings listed or referred to in Engineering Parts Lists (EPL) 8736605 for the M274A2 and (EPL) 8736756 for the M274A5. All parts, assemblies, and subassemblies requiring identification shall be identified in accordance with Standard MIL-STD-130.

3.4.1 Seals. When fording, the seals shall be capable of preventing the entrance of foreign matter into the bearings, which are exposed to contamination during these operations. All bearing seals shall prevent the leaking of lubricants from the bearings.

3.4.2 Fuel system. During vehicle operation, the fuel system shall maintain continuous supply of fuel to engine when vehicle is ascending and descending 60 percent grades in forward and reverse gear, and on 40 percent side slopes, with each side of vehicle up slope.

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3.4.2.1 Fuel tank and lines. The fuel tank, lines, and connections shall be internally clean prior to initial fueling of vehicle and shall be free from leaks.

3.4.3 Engine and power train. The engine conforming to DAPD-298, and power train components shall be installed and adjusted to provide the performance and operational characteristics specified herein.

3.4.3.1 Exhaust system. The exhaust system as installed shall be gas-tight and leak proof to prevent concentration of exhaust gas in the occupied areas and shall be in accordance with HEL-STD-S-6-66.

3.4.4 Steering mechanism. Steering mechanism for the M274A2 shall provide selection of steering by the driver of the 2 front wheels or all 4 wheels at minimum speed, and operational control from a sitting, walking, and crouching position. When using 4 wheel steer, the vehicle shall turn within a circle of 23 feet 4 inches over the left corner, and within a circle of 21 feet 4 inches over the right corner. When using two wheel steer, the vehicle shall turn within a circle of 44 feet 6 inches over the left corner and turn within a circle of 38 feet over the right corner. The steering mechanism for the M274A5 shall control the two front wheels and at operator's selection provide operational control from a sitting, walking, and croutching position. The vehicle shall turn within a circle of 44 feet 6 inches over the left corner and turn within a circle of 40 feet over the right corner.

3.4.5 Lifting devices. Four lifting devices for use with vehicle lifting slings shall be provided. The lifting devices shall have sufficient strength to lift the vehicle with payload. These devices shall be integral with vehicle, and provide a stable attitude for loaded vehicle when lifted to or from shipboard or for helicopter flight and shall be in accordance to MIL-STD-209.

3.4.6 Special kits. When specified (see 6.2) special kits shall be furnished as follows:

3.4.6.1 Deep water fording kit. When installed on the vehicle, the deep water fording kit shall assure vehicle performance as specified herein.

3.4.6.2 Bat weapon adaptation kit. The bat weapon adaptation kit furnished shall conform to applicable drawings. This kit is intended for adapting the 106 mm recoilless rifle (kit).

3.4.6.3 Litter kit. The litter kit furnished shall conform to applicable drawings. This kit shall provide for transporting incapacitated personnel. With driver's seat installed space shall be provided for installation of 3 litters, and with 4 litters when driver's seat is removed and stowed.

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3.5 Break-in run. Each vehicle shall be operated by the contractor for a distance of 5 miles at speed from minimum speed up to maximum speed and through full range of gears to assure vehicle will be acceptable prior to being submitted to the Government for acceptance. A test stand conforming to Drawing C10957514, may be used for 4 miles of the break-in run, but a minimum of 1 mile shall be under actual road conditions.

3.5.1 Preparation. Prior to vehicle run in, the vehicle shall be completely assembled and all adjustments completed. The vehicle shall be lubricated as specified in production lubrication drawing C10944366 a detail of EPL 8736605 or EPL 8736756. The vehicle shall be supplied with fuel conforming to Specification MIL-G-3056. Prior to start of, or any time during vehicle run-in when the engine has been stopped for at least 15 minutes, the engine when restarted shall be operated at fast idle for approximately 2 minutes to allow engine and lubrication warmup.

3.6 Performance conditions and characteristics. Vehicle serviced with Department of the Army Standard Products (see Table I) and with rated payload (see 3.7), shall meet the performance requirements specified herein.

Table I. Standard products

Products	Initial fill only	Specifications for use in ambient air temperature ranges	
		0°F to plus 115°F.	Minus 25°F to 0°F.
Gasoline.....	MIL-G-3056 (type I)	MIL-G-3056 (type I)	MIL-G-3056 (type II)
Engine oil.....	MIL-L-21260 (type I)	MIL-L-2104	MIL-L-2104
Gear lubricant: For general chassis lubrication wheel bearings and universal joints.		MIL-L-2105 MIL-G-10924	MIL-L-10324 MIL-G-10924

3.6.1 Environmental requirements.

3.6.1.1 Temperature conditions. The vehicle shall be capable of starting and demonstrating the performance characteristics specified herein under any of the following conditions or combination of conditions:

(a) Without external aids after being thoroughly cold-soaked at ambient air temperatures down to -25°F.
(b) Without external aids when operated at an ambient air temperature of 125°F.

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3.6.1.2 Limiting operating temperatures. The vehicle, with pay load, shall not exceed the specified limiting temperatures when operating at full throttle, throughout the speed range of the engine under any of the conditions specified herein. The engine operating temperatures shall be in accordance with DAPD-298.

3.6.2 Fording ability.

3.6.2.1 Fording shallow water. Vehicle shall ford hard-bottom water crossings in salt or fresh water, up to 18 inches deep, without the use of modification kit equipment and without special adjustment.

3.6.2.2 Fording deep water. With special equipment installed, the vehicle shall ford up to 48 inches in depth including wave height.

3.6.3 Speeds. Vehicle shall maintain the speeds specified herein, on smooth, dry, level, and hard-surfaced roads.

3.6.3.1 Maximum speed. Maximum speed of vehicle shall be not less than 25 miles per hour (mph) at an engine speed that does not exceed the maximum engine operating speed recommended by the engine manufacturer.

3.6.3.2 Minimum speed. Minimum sustained speed of vehicle shall be not more than 1 mph.

3.6.3.3 Towed speed. Vehicle shall withstand being towed by other military vehicles up to and including 2 1/2 ton, 6 x 6, truck at speeds up to 50 miles per hour.

3.6.4 Climbing.

3.6.4.1 Normal incline. Vehicle shall climb a normal incline of not less than 5 percent at a road speed of not less than 19 mph.

3.6.4.2 Maximum incline. Vehicle shall ascend an incline of not less than 60 percent, fore and aft, at a speed of not more than 2 mph.

3.6.4.3 Side slope. Vehicle shall operate on side slopes, right and left, up to 40 percent.

3.6.5 Braking ability.

3.6.5.1 Stopping. Service brake shall bring the vehicle to a complete stop from a speed of 20 mph, within 30 feet, from position at instant of brake application, on dry, relatively level, smooth road, free from loose material.

3.6.5.2 Service brake. Service brake, when applied, shall stop, hold motionless, and control the vehicle when operating on 60 percent grades fore and aft.

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3.6.5.3 Parking brake. Parking brake, when applied and locked for parking, shall hold the vehicle motionless on 60 percent grades fore and aft.

3.7 Dimensions, weights, and loads. Dimensions, weights, and loads shall conform to Table II.

Table II. Dimensions, weights, and loads

Wheelbase, inches	57
Front Tread, inches	40 1/2
Rear Tread, inches	40 1/2
Curb weight, pounds	970
Payload, cross-country, and highway, pounds	1000
Length, reducible to, inches	103 1/8 ± 1/8
Height, reducible to, inches	28.94
Minimum ground clearance, inches	8 1/2
Angle of approach, degree	40
Angle of departure, degree	34

3.7.1 Curb weight. Curb weight shall include weight of the fully equipped vehicle, including fuels and lubricants.

3.7.2 Payload allowance. Payload allowances specified in table II shall include weight of operator (175 pounds).

3.8 Painting. Unless otherwise specified (see 6.2), painting shall be in accordance with MIL-STD-193, except as required by the using service.

3.9 Marking. Vehicle identification marking shall be in accordance with MIL-STD-642.

3.10 Radio-interference suppression. Vehicle shall be radio interference suppressed in accordance with the tactical vehicle requirements Class IIIA of MIL-STD-461.

3.11 Servicing. Vehicles shall be serviced and all components adjusted for operational use including at least the following: lubrication of vehicle, adjustment of the engine, brake and steering mechanism controls.

3.12 Workmanship. Workmanship shall be of a quality to assure vehicles free of defects resulting from improper manufacturing or assembly practices and conforming to requirements and tests specified herein.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Parts and components. Parts, components and assemblies shall be inspected in accordance with the drawings listed on EPL's 8736605 or 8736756 as applicable and Master List of Supplementary Quality Assurance Provisions (MSQAP) 8736605 or 8736756 as applicable.

4.1.2 Contractor's quality control system. Unless otherwise specified by the procuring activity, the contractor shall provide and maintain an effective inspection and Quality Control System acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality control procedure. The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of control required by this specification or other applicable documents referenced therein.

4.1.3 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an effect upon the quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until corrective action has been made or until conformance to product to prescribed criteria has been demonstrated.

4.1.4 Inspection equipment. Inspection equipment drawings referenced by number herein are details of Master Lists of Inspection Equipment Drawings 8736605 or 8736756. Inspection Equipment Drawings listed thereon will serve as standards for determining the adequacy of contractor's inspection equipment (see 6.3).

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4.1.5 Qualified products. The contractor's inspection records shall be checked to determine contractor conformance to 3.3.1. The contractor shall provide the Government with evidence that materials (see 3.3) used are as specified in the applicable specifications and drawings.

4.2 Preproduction and first production inspection.

4.2.1 Preproduction inspection. A preproduction vehicle (see 3.1) when furnished shall be inspected to ascertain vehicle conformance with the requirements of this specification. Inspection shall be performed at a location approved by the Government.

4.2.2 First production inspection.

4.2.2.1 Special process inspection. A special process inspection will be conducted by the Government representatives during fabrication of the first production vehicle (see 3.2) to evaluate conformance of materials and workmanship with drawing and specification requirements.

4.2.2.1.1 Review and evaluation. Review and evaluation of records for processing procedure, inspection systems and plan of quality control, will be conducted on the following: material, welding, surface hardening, heat treating, protective finishes, etc.

4.2.2.2 Completed first vehicle inspection. The first production vehicle of each model on each contract shall be road tested and completely inspected by the contractor for conformance to the requirements of contract and specifications. The applicable road test shall be in accordance with paragraph 4.5.2 upon completion, the contractor shall submit the vehicle (and make available all inspection records and certifications) to the responsible government inspection element at contractor's plant for preliminary inspection.

4.2.2.2.1 Preliminary inspection. The vehicle shall receive a preliminary inspection by the responsible Government inspection element. The preliminary inspection shall be complete except for road test requirements of paragraph 4.5.2 which will be conducted jointly during provisional inspection to be accomplished by representatives of USATACOM Quality Assurance Directorate as specified in para 4.2.2.2.2.

4.2.2.2.2 Provisional inspection. The first completed production vehicle shall be subject to provisional inspection at the contractor's plant by representatives of USATACOM Quality Assurance Directorate and shall further provide any other inspection assistance as may be required. At the time of this inspection, the contractor shall make available his inspection plan, inspection records and certifications pertinent to the vehicle and components.

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4.2.2.2.3 Corrections. Corrections to the vehicle, found necessary as a result of above inspection shall be made by the contractor.

4.2.2.3 Vehicle disposition. Upon completion of inspection and acceptance of the first production vehicle, the vehicle shall remain at the contractors facility to serve as a manufacturing standard for acceptance criteria and will be shipped upon completion of the production run. The manufacturer shall service and maintain the vehicle during this period in accordance with applicable documents for care and preservation while in storage.

4.2.3 Production vehicle test. After completion of the first vehicle inspection, additional vehicles shall be selected from the first months production and subjected to examinations (see 4.4.2) and tested as specified in 4.5 (a) and (b). Subsequent to examination and test, the selected vehicles shall be subjected to the 4,000 mile test specified in Table III. These test shall be performed by the Government, at a test site(s) approved by the Government. Delays caused by vehicle break-down due to poor workmanship or material, or failure of the contractor to comply with specifications or drawing requirements shall not be the basis for adjustment of the contract, performance dates or delivery schedule or contract price. The contractor will expeditiously furnish repair parts, when required to adequately support the above tests. Delays caused by vehicle break-down and failure of the contractor to adequately support the vehicle with parts during the tests shall not be the responsibility of the Government.

Table III. 4,000 mile test rated payload (Government Proving Ground)

Course	Mileage and speeds	Vehicle loading
Level cross country	1200 miles at variable speeds	Rated payload
Hilly cross country	1200 miles at variable speeds	Rated payload
Gravel and dirt roads	1200 miles at variable speeds	Rated payload
Belgian block	100 miles at variable speeds	Rated payload
Paved roads	300 miles at variable speeds	Rated payload

4.2.3.1 Test failure. Failure of the vehicle as the result of any deficiency of a workmanship or materials nature during or as result of the 4,000 miles, will be cause for rejection of the vehicle. Further, the Government may refuse to continue acceptance of production vehicles until evidence has been provided by the manufacturer that corrective action has been taken to eliminate the deficiency. Any deficiency found during or as a result of 4,000 mile test shall be prima facie evidence that all vehicles already accepted prior to completion of the 4,000 mile test are similarly deficient unless evidence satisfactory to the 4,000 mile test are similarly deficient unless evidence satisfactory to the contracting officer is furnished by the contractor that they are not similarly deficient. Such deficiencies on all vehicles shall be corrected by the contractor at no cost to the Government.

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4.2.3.2 Reliability verification. To determine conformance to 3.2.1, the reliability requirements will be verified at a .90 confidence level while the vehicle is subjected to production vehicle test (see 4.2.3). The mean miles between failure (MMBF) shall not be less than specified.

4.2.3.3 Durability verification. To determine conformance to 3.2.2, the durability requirements will be verified at a .50 confidence level while the vehicle is subjected to production vehicle test (see 4.2.3).

4.2.3.4 Maintainability verification. To determine conformance to 3.2.3, the maintainability requirements will be verified during production vehicle test (see 4.2.3).

4.3 Break-in run. To determine conformance to 3.5 and 3.5.1, each vehicle shall be lubricated and serviced prior to the break-in run. The run shall be on smooth, level, hard-surfaced roads, and for the distance specified.

4.3.1 Simulated break-in run. To determine conformance to 3.5 and 3.5.1, a simulated break-in run may be accomplished on a vehicle run-in test stand in accordance with performance test equipment specifications drawing C10957514 with a minimum of one (1) mile of the total break-in run (5 miles) on smooth, level, hard surfaced roads.

4.4 Inspection provisions.

4.4.1 Lot size. An inspection lot shall consist of all vehicles of one type, from one day's production, submitted at one time for acceptance inspection.

4.4.1.1 Sampling for inspection. For the purpose of visual, dimensional, and primary functional inspection, a representative sample shall be selected, from each inspection lot in accordance with inspection level II of MIL-STD-105. Before sampling may be accomplished by the contractor, the contractor shall 100 percent inspect the first 20 vehicles to establish a process average (see 6.4) to allow normal sampling in accordance with MIL-STD-105.

4.4.2 Examinations. Visual, dimensional, and primary functional examination shall consist of examination of the vehicle for conformance to the applicable drawings and this specification. Examination shall be performed against the classification of defects and with the acceptable quality levels shown in Table IV. The following constitutes a part of the classification of defects (see 6.5):

(a) Any drip constitutes a major defect when the vehicle has been standing idle and the components are at ambient temperature.
(b) Any droplet that occurs at a static fit, metal or gasket combination shall constitute a major defect.

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Table IV.

Area	Characteristics Major: 25/Defects 100 units		Minor: 150 Defects/100 units		Method of inspection
Engine assy. (includes fuel pump, carburetor, governor, magneto, spark plugs, oil pump oil filter, belt cooling impeller	101.	Malfunction; improper mounting; seals defective; leaking or missing; blower damage; improper fan guard clearance; ignition wiring damage.	201.	Improper adjustment or installation of components; improper belt tension.	Visual and functional
Transmission	102.	Malfunction, improper mounting	202.	Defective breather; improper component adjustment.	Visual and functional
Clutch	103.	Malfunction; contamination	203.	Improper adjustment	Visual and functional
Axles and drop gear cases	104.	Malfunction; leaks			Visual and functional
Propeller shaft and universal joints	105.	Misalinement; component damage.	204.	Improper assembly and installation	Visual and functional
Brake system components and controls	106.	Malfunction; cable or quick disconnect damage; brake pedal operation, improper adjustment.			Visual and functional
Steering mechanism	107.	Malfunction of components; improper clearance; damaged components.	205.	Improper adjustment	Visual and functional
Controls and linkage; (includes shift, throttle, choke, clutch, starter mechanism)	108.	Malfunction; component, cable or conduit damage; interference	206.	Improper installation; improper adjustment; defective boots.	Visual and functional

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Table IV. (Cont'd)

Area	Characteristics		Method of inspection
	Major: 25/Defects 100 units	Minor: 150 Defects/100 units	
Fuel system components, fuel lines, hose, vents, shut-off valve	109. Malfunction; damage; leaks; interference; evidence of contamination.	207. Improper assembly or installation of components.	Visual and functional
Exhaust system	110. Exhaust leaks; damaged tubing or muffler.	208. Weld defects; improper or defective installation.	Visual and functional
Frame structure and platform.	111. Weld defects; damage	209. Improper assembly or installation.	Visual
Driver's seat	112. Weld defects; improper mounting; catch damage	210. Upholstry defects; improper installation	Visual and functional
Wheels, tires and tubes	113. Wheel damage; detrimental tubes and tire defects and loose nuts	211. Improper type or size; improper mounting; improper inflation.	Visual and tactile
Miscellaneous items, accessories (includes air cleaner, lifting devices, access covers, rails)	114. Malfunction; missing; damaged; improperly located.		Visual
Painting and marking		212. Improper painting; improper or illegible marking; improper marker location.	Visual

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4.4.3 Unclassified defects. All defects that have no effect on function, safety, interchangeability, or life, but that are considered departures from good workmanship will be noted in writing. Workmanship deficiencies falling within this category and recurring in five consecutive lots, or ten lots or more within a thirty-day period will be added to the minor classification with no increase in AQL's.

4.4.4 Recurring major deficiencies. A major deficiency (see 6.6) is recurring when the same defect occurs more than once in the same sample, or when the defect occurs in two successive samples. A major defect may be considered recurring when the historical inspection records ("P" Chart or approved equivalent) reflect such a condition. Recurring major deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies. The deficiencies shall be corrected when found.

4.4.5 Recurring minor deficiencies. A minor deficiency (see 6.6) is recurring if it occurs more than twice in the same sample, or when the defect occurs in four successive samples. Recurring minor deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies and shall be accomplished prior to acceptance.

4.5 Classification of tests.

(a) Acceptance tests 4.5.1 through 4.5.1.9
(b) Control tests 4.5.2 through 4.5.2.4
(c) Inspection comparison tests (ICT) 4.5.3 through 4.5.3.7

4.5.1 Tests.

4.5.1.1 Acceptance tests. To determine conformance to 3.6, after break-in run (see 3.5), the vehicle selected in accordance with 4.4.1.1 shall be operated for a distance of 5 miles or more and shall be inspected as specified in 4.4.2 and subjected to acceptance tests, which includes all acceptance tests specified at the place of manufacture in table V. Except for equipment and component installation and servicing performed by the contractor as a final processing of vehicles for acceptance, vehicles shall be completely assembled and serviced with fuel, lubricants, and fluid as specified, but need not be loaded with rated payload. After the tests are completed, the vehicle shall be examined for evidence of fuel and lubricant leakage, or other deficiencies. Prior to performing these tests, the engine shall be operated at a fast idle for not less than 2 minutes to allow the engine and lubricant warmup. In lieu of conducting the actual road test, the contractor may determine conformance to the specification by simulating the road test conditions through the use of a 4 wheel drive test stand. The contractor shall develop a detailed test procedure for the test stand, and submit same to the procurement activity for review and approval before testing.

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Table V. Location for tests

Title	Check or Test Paragraph	Place of Manufacturer	Government Proving Ground
Preproduction			
Vehicle test	4.2.1		X
Production vehicle test	4.2.3		X
Reliability test	4.2.3.2		X
Durability test	4.2.3.3		X
Maintainability	4.2.3.4		X
Acceptance tests	4.5.1.1	X	X
Fuel system check	4.5.1.2	X	X
Fuel tank and lines check	4.5.1.3	X	X
Engine and power train check	4.5.1.4	X	X
Exhaust system check	4.5.1.4.1	X	X
Steering mechanism check	4.5.1.5	X	X
Lifting devices check	4.5.1.6	X	X
Special kits check	4.5.1.7	X	X
Speed tests	4.5.1.8	X	X
Stopping test	4.5.1.9	X	X
Control tests	4.5.2	X	X
Maximum and minimum speed tests	4.5.2.2	X	X
Normal incline test	4.5.2.3	X	X
Braking ability test	4.5.2.4	X	X
Inspection comparison tests	4.5.3		X
Special kit test	4.5.3.2		X
Environmental tests	4.5.3.3		X
Fording ability test	4.5.3.4		X
Towed speed test	4.5.3.5		X
Climbing tests	4.5.3.6		X
Radio suppression test	4.5.3.7		X

4.5.1.2 Fuel system check. To determine conformance to 3.4.2, with the vehicle operating as specified, it shall be observed for fuel supply to the engine.

4.5.1.3 Fuel tank and lines check. To determine conformance to 3.4.2.1, the tank and lines shall be checked for leaks.

4.5.1.4 Engine and power train check. To determine conformance to 3.4.3, the engine and power train shall be checked for proper assembly, installation, and adjustments.

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4.5.1.4.1 Exhaust system check. To determine conformance to 3.4.3.1, the exhaust system shall be checked for proper installation and leakage.

4.5.1.5 Steering mechanism check. To determine conformance to 3.4.4, the steering mechanism shall be checked for proper assembly, installation, mounting, turning circle requirements, and operated from a sitting, walking and crouching position.

4.5.1.6 Lifting devices check. To determine conformance to 3.4.5, the lifting devices shall be checked for proper type and installation.

4.5.1.7 Special kits check. To determine conformance to 3.4.6 through 3.4.6.3, when a specific kit is furnished, it shall be checked for completeness and assembly.

4.5.1.8 Speed tests. To determine conformance to 3.6.3 through 3.6.3.3, the vehicle shall be operated as specified and checked for meeting performance requirements.

4.5.1.9 Stopping test. To determine conformance to 3.6.5.1, the vehicle shall be operated as specified, checked for meeting stopping requirements.

4.5.2 Control tests. The Government shall select at random for road test by the contractor, one per month or one out of each 100 vehicles submitted, but not more than 2 vehicles in any given 30-day period. The vehicle shall be operated for a distance of not less than 50 miles with the rated payload or a simulated payload and all equipment installed and shall be subjected to the tests specified in Table V at place of manufacture.

4.5.2.1 Test failure. If a vehicle fails to pass any control test specified herein, the Government shall stop acceptance of subsequent lots until objective evidence has been provided by the manufacturer that corrective action has been taken to eliminate the condition which cause rejection.

4.5.2.2 Maximum and minimum speed test. To determine conformance to 3.6.3.1 and 3.6.3.2, the vehicle selected shall be operated as specified and checked for performance requirements.

4.5.2.3 Normal incline test. To determine conformance to 3.6.4.1, the selected vehicle shall be operated as specified and observed for performance requirements.

4.5.2.4 Braking ability tests. To determine conformance to 3.6.5.1 through 3.6.5.3, the selected vehicle shall be operated as specified and checked for meeting the braking requirements.

4.5.3 Inspection comparison tests. The Government may select vehicles any time during the contract period, and subject vehicles to tests specified

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Table V and VI. Test shall be conducted by the Government at a site selected or approved by the Government. Vehicle selected shall not include any vehicles previously tested for conformance to 4.5.2 (control tests).

Table VI. 4,000 mile test rated payload (Government Proving Grounds)

Course	Mileage and speeds	Vehicle loading
Level cross country	1200 miles at variable speeds	Rated payload
Hilly cross country	1200 miles at variable speeds	Rated payload
Gravel and dirt roads	1200 miles at variable speeds	Rated payload
Belgian block	100 miles at variable speeds	Rated payload
Paved roads	300 miles at variable speeds	Rated payload

4.5.3.1 Test failure. Failure of any vehicle to comply with any of the requirements specified in the contract or any major deficiency of a workmanship or materials nature occurring during or as a result of the test cycle, may be considered cause for refusal to continue acceptance of vehicles by the Government until evidence has been provided by the manufacturer that corrective action has been taken by the manufacturer to eliminate the conditions which caused rejection.

4.5.3.2 Special kit test. To determine conformance to 3.4.6 through 3.4.6.3, when a kit is furnished it shall be mounted or installed on vehicle and checked for assurance that it shall meet performance requirements.

4.5.3.3 Environmental tests. To determine conformance to 3.6.1.1 and 3.6.1.2, the vehicle shall be operated under specified conditions, or by means that would assure vehicle will meet the specified requirements.

4.5.3.4 Fording ability test. To determine conformance to 3.4.1 and 3.6.2 through 3.6.2.2, the vehicle with special equipment attached shall be operated as specified and checked for meeting the performance requirements. After fording test, all wheels shall be removed to assure that lubricants are not excessively contaminated.

4.5.3.5 Towed speed test. To determine conformance to 3.6.3.3, the selected vehicle shall be towed by the specific vehicle and observed for towed ability.

4.5.3.6 Climbing tests. To determine conformance to 3.6.4.1 through 3.6.4.3, the vehicle shall be operated on specified grades and side slopes, and observed for performance requirements.

4.5.3.7 Radio interference suppression test. To determine conformance to 3.10, the vehicle shall be subjected to a radio interference suppression

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4.6 Preservation, packaging and vehicle processing. Preservation and packaging of vehicle shall be inspected prior to shipment to assure conformance with the applicable requirements specified herein.

5. PREPARATION FOR DELIVERY

5.1 Vehicle processing. Vehicles shall be processed for shipment and storage in accordance with Standard MIL-STD-281 to the extent indicated on the Vehicle Preservation Data Sheets or other implementing document, as furnished by the responsible Government activity at the time of invitation for bid (see 6.2).

6. NOTES

6.1 Intended use. The vehicles covered by this specification are intended for general use by the United States Armed Services transporting weapons or cargo during tactical military operations.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Vehicle classification as required (see 1.2).
(c) Preproduction model if required (see 3.1).
(d) Special kits, if required (see 3.4.6).
(e) Vehicle color as required (see 3.8).
(f) Selection of applicable levels of preservation (see 5.1).

6.3 Inspection equipment calibration and inspection. Inspecting the supplier's inspection equipment, the Government inspector will determine that the supplier has available, and utilizes correctly, gaging, measuring, and test equipment of the required accuracy and precision. Except as otherwise specified, the supplier will have available a set of master gages, standards, and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration will be maintained by the supplier and made available for review by the Government. The calibration of gages, standards, and instruments will be periodically checked by authorized Government personnel.

6.4 Process average. Sampling may be initiated if the process average value for the first twenty vehicles inspected is less than the AQL specified in the classification of defects for major and minor defects.

$$\text{Process Average} = \frac{\text{Number of defects}}{\text{Number of vehicles inspected}} \times 100$$

If the computed process average exceeds the specified AQL, 100% inspection shall be performed and continues until such time that the process average for twenty consecutive vehicles is less than the specified AQL.

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6.5 The following shall be used with the classification of defects:

(a)	Weep:	Any evidence of fluid beyond the seal.
(b)	Seep:	Any evidence of fluid beyond the seal that does not result in function of a droplet.
(c)	Droplet:	Any evidence of fluid beyond the seal that results in the formation of a droplet.
(d)	Drip:	Any evidence of fluid beyond the seal where droplet forms and falls.

6.6 Definitions of recurring major and minor defects. The Government Inspector will verify that a thorough inspection of each vehicle is performed by the contractor not only for the listed characteristics, but also for any other departures from good workmanship. The Government Inspector will assure that all deficiencies encountered during the inspection are enumerated on the deficiency sheet for the vehicle. The defects noted on Deficiency Sheet shall contain sufficient description to enable the Government Inspector and the contractor's representative to classify the deficiency in accordance with the classification of defects of the vehicle specification and definitions contained in MIL-STD-105. Corrective action shall be taken for recurring deficiencies.

6.7 Engine exhaust and crankcase emissions.

(a) Emission levels from the engine shall comply with the Federal requirements for Heavy Duty Gasoline Engines as specified in Volume 35, Number 219 of the Federal Register and the engine shall be covered by a "certificate of conformity" awarded by the Environmental Protection Agency.

(b) The vehicle and the vehicle's engine shall be modified or changed to assure compliance with any amendments or changes to the Federal Law which affect mission standards for the vehicle and the vehicle's engine. The vehicle and the vehicle's engine shall be covered by a "certificate of conformity" for any new amendments or changes to the Federal Law concerning emissions standards.

(c) The vehicle with emission control shall demonstrate compliance with the performance and other capabilities specified herein.

Custodian:
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Navy - MC

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Army - AT

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